

GPA Calculator Web Application SEN210

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RESEARCH PROBLEM

Many people struggle to manage their monthly income and expenses effectively.

Traditional budgeting methods such as manual calculations or paper tracking can be inconvenient and error-prone.

Users need a simple and interactive system that can:

- ▶ Track income
- ▶ Record expenses
- ▶ Calculate remaining balance automatically
- ▶ Display financial information clearly

This project aims to solve these issues through a desktop-based budget tracking application.

MOTIVATION

The motivation behind this project was to create a user-friendly financial management tool using Python and Qt.

Control Flow

- ▶ QApplication object is created
- ▶ Main BudgetApp window is initialized
- ▶ User enters monthly income
- ▶ User clicks **Set Income**
- ▶ `set_income()` function executes
- ▶ `add_expense()` function executes
- ▶ Remaining balance recalculates

- ▶ Object-Oriented Design
- ▶ The project uses Object-Oriented Programming through the BudgetApp class.

▶ UI Design Strategy

- ▶ The interface was designed using:
- ▶ QVBoxLayout for vertical organization
- ▶ styled widgets using Qt stylesheets

IMPLEMENTATION STRATEGY